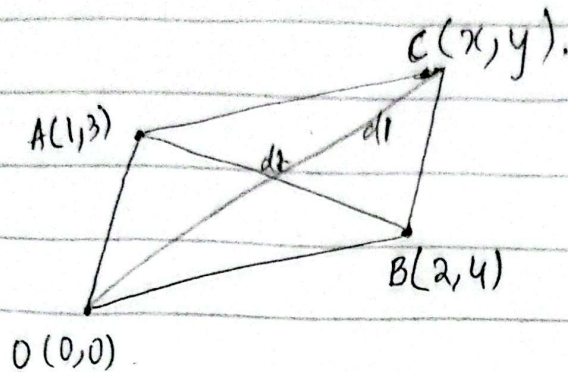


Assignment-1 "Multivariable Calculus."

Q1



$$d_1 = d_2$$

Sum of coordinates of d1 = Sum of coordinates of d2

$$O(x_0) + C(x_c) = A(x_A) + B(x_B)$$

$$0 + x = 1 + 2,$$

$$x = 3.$$

$$O(y_0) + C(y_c) = A(y_A) + B(y_B)$$

$$0 + y = 3 + 4$$

$$y = 7.$$

Fourth vertex of $\square OACB = (3, 7)$.

$$C = (3, 7)$$

Q2

a. Parallel to the line $y = 3x + 2$.

Finding the direction vector $v = ?$

$$m = \frac{\text{rise}}{\text{run}} = \frac{3}{1}$$

$$V = \langle v_{\text{run}}, v_{\text{rise}} \rangle = \langle 1, 3 \rangle$$

$$\text{vector}_1 = \frac{V}{\|V\|} = \frac{\langle 1, 3 \rangle}{\sqrt{(1)^2 + (3)^2}} = \frac{\langle 1, 3 \rangle}{\sqrt{10}}$$

$$= \left\langle \frac{1}{\sqrt{10}}, \frac{3}{\sqrt{10}} \right\rangle$$

$$\text{vector}_2 = \frac{-V}{\| -V \|} = \frac{\langle -1, -3 \rangle}{\sqrt{(-1)^2 + (-3)^2}} = \frac{\langle -1, -3 \rangle}{\sqrt{10}}$$

$$= \left\langle \frac{-1}{\sqrt{10}}, \frac{-3}{\sqrt{10}} \right\rangle$$

b. Parallel to the line $x + y = 4$

$$y = 4 - x$$

$$m = -1$$

$$V = \langle 1, -1 \rangle$$

$$\text{vector}_1 = \frac{\langle 1, -1 \rangle}{\sqrt{(1)^2 + (-1)^2}} = \frac{\langle 1, -1 \rangle}{\sqrt{2}} = \left\langle \frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}} \right\rangle$$

$$\text{vector}_2 = \frac{-V}{\| -V \|} = \frac{\langle -1, 1 \rangle}{\sqrt{(-1)^2 + (1)^2}} = \frac{\langle -1, 1 \rangle}{\sqrt{2}}$$

$$= \left\langle \frac{-1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right\rangle$$

c. Perpendicular to the line $y = -5x + 1$

$$m = -5$$

$$m \times m_{\perp} = -1$$

$$m_{\perp} = \frac{-1}{-5} = \frac{1}{5}$$

$$V = \langle 5, 1 \rangle$$

$$\text{vector}_1 = \frac{V}{\|V\|} = \frac{\langle 5, 1 \rangle}{\sqrt{(5)^2 + (1)^2}} = \frac{\langle 5, 1 \rangle}{\sqrt{26}} = \left\langle \frac{5}{\sqrt{26}}, \frac{1}{\sqrt{26}} \right\rangle$$

$$\text{vector}_2 = \frac{-V}{\| -V \|} = \frac{\langle -5, -1 \rangle}{\sqrt{(-5)^2 + (-1)^2}} = \frac{\langle -5, -1 \rangle}{\sqrt{26}} = \left\langle \frac{-5}{\sqrt{26}}, \frac{-1}{\sqrt{26}} \right\rangle$$

Q3 a. $A(2, -1, 1)$ $B(3, 2, -1)$ $C(7, 0, -2)$



$$\vec{BA} = \langle 3-2, 2-(-1), -1-1 \rangle$$

$$= \langle 1, 3, -2 \rangle$$

$$\vec{BC} = \langle 7-3, 0-2, -2-(-1) \rangle$$

$$= \langle 4, -2, -1 \rangle$$

$$\vec{BA} \cdot \vec{BC} = 0$$

$$4 + (-6) + 2 = 0$$

$$4 + 2 - 6 = 0$$

$$0 = 0$$

Hence, these vertices are

of a right triangle.

Also the right angle is at vertex B as

$$\vec{BA} \cdot \vec{BC} = 0$$

whereas $\vec{AC} \cdot \vec{AB} \neq 0$ & $\vec{CA} \cdot \vec{CB} \neq 0$

$$\vec{AC} \cdot \vec{AB} \neq 0$$

$$\vec{CA} \cdot \vec{CB} \neq 0$$

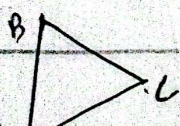
$$5 + 3 + 6 \neq 0$$

$$20 + (-2) + 3 \neq 0$$

$$14 \neq 0$$

$$21 \neq 0$$

b. $A(-1, 0)$ $B(2, -1)$ $C(1, 4)$



$$\vec{AB} = \langle 3, -1 \rangle$$

$$\vec{AC} = \langle 2, 4 \rangle$$

$$\vec{BA} = \langle -3, 1 \rangle$$

$$\vec{BC} = \langle -1, 5 \rangle$$

$$\vec{CA} = \langle -2, -4 \rangle$$

$$\vec{CB} = \langle 1, -5 \rangle$$

Angle at A.

$$\vec{AB} \cdot \vec{AC} = |\vec{AB}| |\vec{AC}| \cos \theta$$

$$(3 \times 2) + (-1 \times 4) = \sqrt{(3)^2 + (-1)^2} \sqrt{(2)^2 + (-4)^2} \cos \theta$$

$$6 - 4 = \sqrt{10} \sqrt{20} \cos \theta$$

$$\frac{2}{\sqrt{10} \sqrt{20}} = \cos \theta$$

$$\cos^{-1} \left(\frac{2}{\sqrt{10} \sqrt{20}} \right) = \theta_A$$

$$81.9 = \theta_A$$

$$82 \approx \theta_A$$

Angle at B.

$$\vec{BA} \cdot \vec{BC} = |\vec{BA}| |\vec{BC}| \cos \theta$$

$$3 + 5 = \sqrt{9+1} \sqrt{1+25} \cos \theta$$

$$\cos^{-1} \left(\frac{8}{\sqrt{10} \sqrt{26}} \right) = \theta_B$$

$$60.3 = \theta_B$$

$$60 \approx \theta_B$$

Angle at C

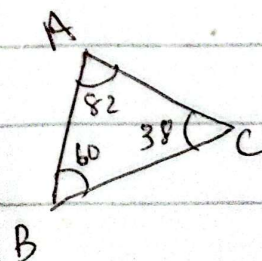
$$\vec{CA} \cdot \vec{CB} = |\vec{CA}| |\vec{CB}| \cos \theta$$

$$-2 + 20 = \sqrt{4+16} \sqrt{1+25} \cos \theta$$

$$\cos^{-1} \left(\frac{18}{\sqrt{20} \sqrt{26}} \right) = \theta_C$$

$$37.9 = \theta_C$$

$$38 \approx \theta_C$$



Q4

$$r_0 = \langle -1, 2 \rangle$$

$$v = \langle 1, 1 \rangle$$

$$r_0 + v = \langle -1+1, 2+1 \rangle = \langle 0, 3 \rangle$$

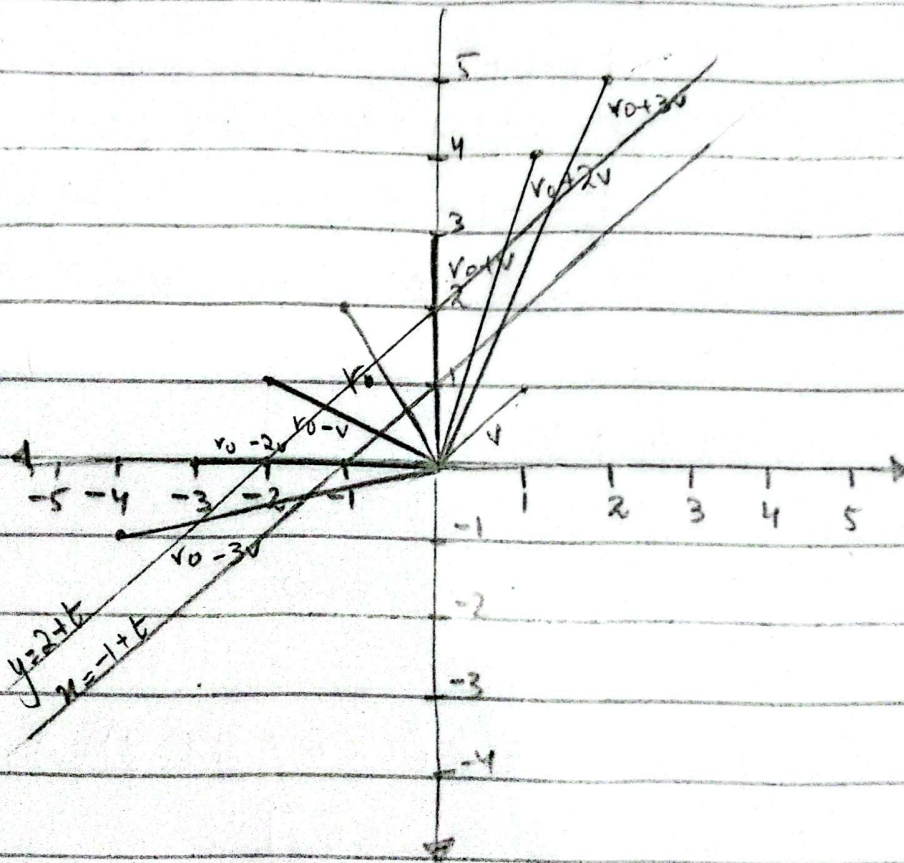
$$r_0 - v = \langle -1-1, 2-1 \rangle = \langle -2, 1 \rangle$$

$$r_0 + 2v = \langle -1+2(1), 2+2(1) \rangle = \langle 1, 4 \rangle$$

$$r_0 - 2v = \langle -1-2(1), 2-2(1) \rangle = \langle -3, 0 \rangle$$

$$r_0 + 3v = \langle -1+3(1), 2+3(1) \rangle = \langle 2, 5 \rangle$$

$$r_0 - 3v = \langle -1-3(1), 2-3(1) \rangle = \langle -4, -1 \rangle$$



$$\text{Line } L : x = -1 + t$$

$$y = 2 + t$$

Relationship b/w line L & the vectors.

• The points $r_0 \pm nv$ (for $n = 1, 2, 3$) all lie on this line.

• Each vector represents a position vector to a point on L .

• As t varies over all real numbers, $r = r_0 + tv$ traces the entire line.

Vector equation of L .

$$r = r_0 + tv$$

$$r = \langle -1, 2 \rangle + t \langle 1, 1 \rangle$$

$$r = -1 + t \Rightarrow x = -1 + t$$

$$r = 2 + t \Rightarrow y = 2 + t$$

Q5

point $(-1, 4, 2)$

Planes $4x - y + z - 2 = 0$

$$2x + y - 2z - 3 = 0$$

$$(4x - y + z - 2) + k(2x + y - 2z - 3) = 0$$

Substituted $(-1, 4, 2)$

$$(4(-1) - (4) + 2 - 2) + k(2(-1) + 4 - 2(2) - 3) = 0$$

$$(-4 - 4 + 2 - 2) + k(-2 + 4 - 4 - 3) = 0$$

$$(-8) + k(-5) = 0$$

$$-5k = +8$$

$$k = \frac{8}{-5}$$

$$\underline{\underline{-5}}$$

$$(4x - y + z - 2) - \frac{8}{5} (2x + y - 2z - 3) = 0$$

$$5(4x - y + z - 2) - 16x - 8y + 16z - 24 = 0$$

$$20x - 5y + 5z - 10 - 16x - 8y + 16z - 24 = 0$$

$$4x - 13y + 21z + 14 = 0$$

Equation of the plane